Project title: Predicting How SJSU Students Commute: Analysis of Machine Learning Accuracy and Fairness

Team information (include names and SJSU IDs):

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Data Set:

The plan is to use publicly available student transportation and commute data from San Jose State University's Annual Student Commute Survey, which are conducted annually every fall semester. The data will include information about students' transportation choices and commuting characteristics. Since we are currently in the fall semester of '26, we will be using the most recently released data.

Note: We are considering dataset reports like the SJSU Student Commute Survey Report–2024, for example. However, we need to determine if the results are available as individual rows of data or just published/aggregated results. For this circumstance, we still plan to use SJSU's Annual Student Commute Survey, but it is subject to availability of an appropriate anonymized dataset.

Project Idea:

SJSU is primarily a commuter campus with diverse student demographics facing different transportation needs based on factors such as commute distance, housing situation, financial means, and access to transportation. We are investigating whether a machine learning model can accurately predict students' primary modes of transportation and whether the model performs differently across different demographics of groups of students. We will investigate various transportation choices such as driving, public transportation, walking, biking, and carpooling. We will also investigate if the model displays any biases and disparities across various student populations. It is vital to investigate whether a predictive model is able to perform at the same level of accuracy across different groups of students.

Approach to solve:

Receive SJSU Commute Data → Clean & preprocess data → Train ML models ↓

Test fairness ← Eval. student group performance ← Measure accuracy, precision, etc.

↓ Compare accuracy vs. Test fairness → Draw conclusions

A Brief Description of the Steps you will take to complete the project.

1) Clean and preprocess the dataset and select appropriate features to keep based on relevance.
2) Train several algorithms. Examples include decision trees and random forests.
3) Evaluate the models on accuracy, precision, recall and F1 score. We are aware that these evaluation metrics are just an overview, but may be refined/edited as data clean develops.
4) Analyze model performance across different student groups to investigate whether removing demographic or potentially sensitive features changes the model's accuracy and fairness.

Expected Results:

We expect to develop strong comparisons of model performance using tables, confusion matrices, and graphs to demonstrate performance across different groups. The tradeoff between the overall predictive accuracy and fairness will be analyzed (Subject to further analysis conditions as the project progresses.

A list of 2 - 3 related references:

1) SJSU Transportation Solutions
2) Student Commute Survey Report 2024

Tentative roles of each team member:

*Still trying to decide roles

[Name]: Analysis on fairness, research, and documentation | Data collection and preprocessing

[Name]: Analysis on fairness, research, and documentation | ML implementation and evaluation